

# Embeddings in 20 Minutes: Turning Meaning into Numbers

High school AI club recap handout

**Big idea**

Embeddings turn meaning into numbers. Once words, phrases, or documents are represented as vectors, AI can compare them by direction and closeness instead of only matching exact keywords.

## What the meeting covered

- Why keyword search can fail. The slide example "jaguar speed" could mean the animal or the car, so matching the same word is not enough.
- What an embedding is: a list of numbers that represents meaning, described in the deck as "like GPS coordinates, but for ideas."
- What a vector is: a list of numbers you can picture as an arrow. The lesson used [3, 2] as a simple 2D example.
- Why cosine similarity matters. The deck emphasized that direction is usually more useful than raw length when comparing meanings.

## Core vocabulary

| Term              | Plain-English meaning                               | Why it matters                                          |
|-------------------|-----------------------------------------------------|---------------------------------------------------------|
| Embedding         | A numeric representation of meaning.                | Lets AI compare ideas mathematically.                   |
| Vector            | A list of numbers that can be drawn as an arrow.    | Gives us a geometric way to reason about meaning.       |
| Cosine similarity | A score for how aligned two vectors are.            | High score means similar direction, so similar meaning. |
| Feature           | A property captured by one coordinate or direction. | Helps explain what the numbers are describing.          |

**Formula idea**

The deck gave cosine similarity as  $\text{cosine}(u, v) = (u \cdot v) / (|u||v|)$ . The point was not to memorize notation. The point was to notice that this score rewards vectors pointing in the same direction and penalizes vectors pointing in opposite directions.

## Worked activity recap

The lesson asked students to compare  $q = (1, 1)$  with  $A = (2, 0)$ ,  $B = (1, 2)$ ,  $C = (-1, -1)$ , and  $D = (3, 3)$ . Using the dot product reminder from the slide, the results are:

| Vector         | q·vector | What it tells us                               |
|----------------|----------|------------------------------------------------|
| $A = (2, 0)$   | 2        | Some similarity, but not as aligned as B or D. |
| $B = (1, 2)$   | 3        | More aligned with q than A.                    |
| $C = (-1, -1)$ | -2       | Opposite direction, so poor match.             |
| $D = (3, 3)$   | 6        | Same direction as q, so best match.            |

### Correct ranking

Most similar to least similar: D, B, A, C. D points in exactly the same direction as q. C points the opposite way.

## Where embeddings show up in real life

- Semantic search: find documents by meaning, not just exact keywords.
- Recommendations: suggest items similar to what you already liked.
- RAG systems: retrieve relevant documents first, then answer with that context.

## What to remember

- Embeddings are a way to encode meaning numerically.
- Similarity often depends more on direction than on vector length.
- This is why AI systems can group related ideas even when the exact words differ.

### Exit question from the meeting

Why is direction more useful than length? A good short answer: length can change for many reasons, but direction better captures what kind of thing the vector represents.

Prepared from the AI Club slide deck for recap use.